

AARON CHANG KAR YOONG

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Accomplished Software Engineer and Full-Stack Developer with expertise in designing and deploying production-grade web and mobile applications. Proficient in modern web technologies including React, Laravel, and Flutter with proven experience integrating third-party APIs, implementing CI/CD pipelines, and optimizing database performance. Foundational knowledge in AI/Machine Learning integration and cross-platform mobile development. Seeking opportunities to contribute technical expertise to innovative software projects while continuously expanding knowledge in emerging technologies.

EDUCATION

Bachelor of Computer Science (Major in Software Engineering)

Quest International University, Ipoh, Perak

- Duration: July 2023 – December 2025
- CGPA: 3.84/4.0

Diploma in Information Technology

Quest International University, Ipoh, Perak

- Duration: October 2020 – April 2023
- CGPA: 3.82/4.0

TECHNICAL SKILLS

Programming Languages: HTML/CSS, JavaScript, TypeScript, Dart, PHP, Python, Java, C/C++, MySQL, VB.NET

Frameworks & Libraries: React, React Native, Next.js, Laravel, Express.js, Node.js, Bootstrap 5, jQuery, GSAP, Three.js, Flutter, Expo SDK 54, ML Kit, Lottie, NetInfo, KoolReport

AI/Machine Learning, Computer Vision: TensorFlow, ML Kit, Open AI, OpenCV

Developer Tools & Platforms: Git, GitHub, Laravel Cloud, Firebase, Railway, Ngrok, Laravel Telescope, Visual Studio Code, Thunder Client, CI/CD, Pub.dev packages, Imgbb API, Android Studio

Third-Party Integration: WhatsApp Business API, Instagram API, Payment Gateway, Google Map API, Google Analytics API

EXPERIENCE

Software Engineer (Degree Intern) | Polyware Sdn Bhd July 2025 – Dec 2025 | Ipoh, Perak

- Developed a React Native (Expo & TypeScript) warehouse app, integrating Bluetooth scanning for real-time inventory tracking and ERP synchronization.
- Built and optimized Laravel REST APIs with MySQL, improving query performance for product searches and stock updates.
- Generated internal operational and analytical reports using KoolReport to support decision-making.
- Collaborated with senior developers and warehouse staff to gather requirements, troubleshoot issues, and optimize features based on real-world warehouse operations and business processes.

Full-Stack Developer (Freelance) | March 2025 – Present | Hybrid

- Designed and developed full-stack web applications using Laravel and React, integrating multiple third-party APIs (WhatsApp Business API, Instagram API, payment gateways) for real-time communications and transactions.
- Developed a high-performance fashion landing page using GSAP animations and responsive design principles.
- Integrated multiple third-party APIs (WhatsApp, Instagram, Gmail, Google) to enable real-time communication and automation.
- Managed CI/CD pipeline configuration, SSL implementation, and hosting infrastructure.
- Migrated application storage to Cloudflare R2 (Object Storage) and integrated a CDN, successfully decoupling assets for stateless deployments

Software Engineer (Diploma Intern) | Polyware Sdn Bhd Jan 2023 – April 2023 | Ipoh, Perak

- Designed user-friendly UI/UX for a POS system, adhering to SDLC methodologies.
- Implemented secure authentication logic to ensure strict access control.
- Authored comprehensive technical documentation covering system architecture and component integration.
- Collaborated with development team to integrate functional components based on system requirements

PROJECT

GPT-5 Smart Customer Service Bot

- Developed a real-time chatbot that receives messages and responds intelligently using GPT-5 Mini.
 - Integrated Meta Webhooks for message events, media handling, and verification flow.
 - Configured context-aware system prompts with website data for personalized customer support.
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Warehouse Management Application

- Developed a cross-platform React mobile app using Expo SDK 54 and TypeScript for real-time warehouse operations including QR code scanning, inventory tracking, and production workflow management.
- Integrated Laravel API with JWT authentication, AsyncStorage for session persistence, and NetInfo for network monitoring with offline handling and automatic token refresh.
- Built REST APIs, optimized SQL queries, and used Laravel Telescope for debugging and performance improvements.
- Optimized raw SQL queries for faster product search and stock update operations.

Full-Stack Web Application

- Developed a full-stack Laravel web application with WhatsApp Bot API, Instagram integration, payment gateway and real-time communications.
- Deployed on Laravel Cloud with SSL/HTTPS, configured CI/CD pipeline for automated testing and deployment, implemented database optimization.
- Built responsive frontend using Bootstrap 5, GSAP animations, and Three.js 3D elements.
- Migrated application storage to Cloudflare R2 (Object Storage) and integrated a CDN

QIU Mobile Application

- Developed an AI-powered gallery app using Flutter and Firebase for engineering students to streamline digital content management with Circle-to-Search, text recognition, and entity extraction features.
- Integrated ML Kit and TensorFlow for real-time image analysis and computer vision.
- Built optical character recognition (OCR) functionality for text extraction
- Configured Firebase for secure data storage and authentication.

POS System Development

- Developed POS system following SDLC methodology
- Designed and developed user-friendly and responsive UI/UX
- Implemented secure login credential logic with proper authentication and access control
- Created detailed documentation covering system architecture and component integration

Mobile Appointment Booking System

- Built mobile application enabling users to schedule appointments efficiently
 - Provided flexibility for users to choose preferred doctor, time, and location
 - Integrated single platform for streamlined appointment booking
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Job Market Automated Scraper

- Implemented web scraping logic using BeautifulSoup and Requests to parse HTML and extract job data from career websites
- Integrated Discord Webhooks API to deliver real-time push notifications when target job opportunities are identified
- Developed task scheduling using the Schedule library to execute scraping at configurable intervals (every minute to daily)
- GitHub Repository: <https://github.com/Aaron0879/job-post-notifier-bot>

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Flutter and OCR Course

Udemy

- Issued: July 2024
- Credential ID: UC-ecb11e78-b5ce-46ff-a874-c2db485fd303
- Skills: Flutter, Machine Learning

Object-Based Search

Udemy

- Issued: July 2024
- Credential ID: UC-ef9f8e6b-af96-4226-9f34-26b970c568fe
- Skills: Artificial Intelligence (AI), Flutter

ROS2 Self Driving Car with Deep Learning and Computer Vision

Udemy

- Issued: July 2024
- Credential ID: UC-fb541122-6b44-41ca-8a04-bdd05ae8ae7a
- Skills: Computer Vision, Python (Programming Language), TensorFlow, Neural Networks, OpenCV

LANGUAGES

- English (Fluent)
 - Mandarin Chinese (Native/Fluent)
 - Malay (Conversational)
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